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构石

A Quantitative Study on the Relationship Between Chihaya Anon' s Sweetness and Tang-Laugh Intensity

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ABSTRACT / 摘要 —Chihaya Anon, a core character in the cross-media franchise *BanG Dream! It's MyGO!!!!!*, has developed a distinctive fan-culture phenomenon around her iconic "Tang-laugh" expression. This paper formalizes "sweetness" (Sweetness Index, S) and "Tang-laugh intensity"(Tang-Laugh Intensity, T) within a quantitative framework, combining a two-player zero-sum game payoff matrix with robust statistical modeling. Using simulated quantitative data filtered by the Median Absolute Deviation (MAD), we analyze 50 highlight clips. The results indicate a very strong positive correlation ($r = 0.89$, $P < 0.001$); under a mixed-strategy Nash equilibrium, fans' defensive thresholds and Tang-laugh "attack" strengths exhibit dynamic counterbalancing; and the linear regression model shows significant explanatory power:

$$S = 0.94T + 0.82,$$
$$R^2 = 0.79, \quad F = 178.6, \quad P < 0.001.$$

INDEX TERMS / 关键词 Chihaya Anon; Sweetness Index; Tang-Laugh Intensity; attack–defense game; robust statistics; MAD filtering; fan culture; BanG Dream!

IMPACT STATEMENT / 影响声明 This study translates the ACG subcultural concepts of "sweetness" and "Tang-laugh" into quantifiable attack–defense game indicators. By leveraging simulated data and robust statistics, it reveals a dynamic stability mechanism of affective expression in virtual characters. The findings can inform affect optimization in ACG content production, recommendation strategy design, and cross-cultural research on emotional contagion in fan communities.

I. INTRODUCTION / INTRODUCTION

Chihaya Anon, a first-year student at Haneoka Girls' Academy and the guitarist of the band MyGO!!!!!, has formed a high-stickiness meme culture on platforms such as Bilibili and Douyin through her pink-themed character design and sudden bursts of sweet laughter. "Sweetness" has become a key indicator for perceived cuteness and affective output, while "Tang-laugh" refers specifically to her sudden, highly infectious smile/laugh. This phenomenon has been widely remixed into sound packs and has evolved into a recognizable cultural symbol within ACG communities.

Existing discussions largely remain qualitative (Xiaoming Wang et al., 2024; Moegirlpedia, 2025) and lack quantitative linkage and a game-theoretic interpretation. We construct a two-player zero-sum "defender–attacker" model (defender = fans' psychological barrier; attacker = character' s Tang-laugh expression) and propose:

H1 Sweetness and Tang-laugh intensity have a very strong positive correlation;

H2 The game admits a mixed-strategy Nash equilibrium, with no universally optimal pure strategy;

H3 Different fan groups (age, occupation, scenario) exhibit robust differences in perceived attack–defense dynamics.

II. MATERIALS AND METHODS / MATERIALS AND METHODS

A. Notation

To ensure rigor and reproducibility, we define the key variables and statistical quantities as follows:

S Sweetness level, $S \in [0, 10]$, computed as a weighted score of cuteness, perceived sweetness, and emotional resonance;

T Tang-laugh intensity, $T \in [0, 10]$, computed as a weighted score of laugh duration (s), peak loudness (dB), eye curvature (°), and mouth-corner lift angle (°);

r Pearson correlation coefficient;

β_0, β_1 Intercept and slope in linear regression;

R^2 Coefficient of determination;

MAD Median absolute deviation: $\mathrm{MAD}(x) := \mathrm{median}(|x_i - \tilde{x}|)$, where $\tilde{x} := \mathrm{median}(x)$.

B. Quantifying Tang-Laugh Intensity (Additional Formula)

Let the feature vector be

$$\mathbf{x} = (t,\, L_p,\, \theta_e,\, \theta_m),$$

where t is laugh duration (s), L_p is peak loudness (dB), and θ_e, θ_m are facial angles (°). After linearly normalizing each dimension to $[0, 10]$, we define:

$$T = \sum_{k=1}^{4} w_k x_k, \quad w_k \geq 0, \quad \sum_{k=1}^{4} w_k = 1. \tag{1}$$

C. Study Materials

We sample from the full 13-episode anime *BanG Dream! It's MyGO!!!!!* as well as official PVs and livestream-related derivative content featuring Chihaya Anon. Using stratified random sampling, we select 50 highlight clips (total duration: 210 minutes) spanning daily interactions, band practice, and climax performances to ensure representativeness.

表 I: Two-player Zero-sum Defender–Attacker Payoff Matrix

	Strong attack ($A \geq 8$)	Weak attack ($A < 8$)
High defense ($D \geq 7$)	(0, -1)	(1, 0)
Low defense ($D < 7$)	(-1, 1)	(0, 0)

D. Data Collection and Robust Filtering

We use Adobe Premiere Pro and an OpenCV-based facial-expression module to extract Tang-laugh segments and record duration, frequency, and expression parameters. Fifty active fans (ages 18–30, each with over 100 hours of viewing) provide double-blind Likert (10-point) ratings, from which we compute mean S and T. Outliers are removed via MAD filtering (filter rate: 4.2%).

For any rating sequence $\{x_i\}_{i=1}^{n}$, we use the robust z-score:

$$z_i^{\mathrm{MAD}} = \frac{x_i - \tilde{x}}{1.4826\ \mathrm{MAD}(x)}, \tag{2}$$

and remove observations with $|z_i^{\mathrm{MAD}}| > 3$.

E. Game-Theoretic Model

We construct a two-player zero-sum "defender–attacker" payoff matrix (Table I). Defender strategies: high defense ($D \geq 7$) / low defense ($D < 7$); attacker strategies: strong attack ($A \geq 8$) / weak attack ($A < 8$).

F. Statistical Analysis

We use Python (pandas, scipy, statsmodels) for descriptive statistics, Pearson correlation, simple linear regression, and robust statistical testing. The significance level is 95%, and Cronbach' s $\alpha = 0.94$.

III. RESULTS / RESULTS

Descriptive statistics: mean sweetness $\bar{S} = 7.48$ (MAD = 1.12), mean Tang-laugh intensity $\bar{T} = 7.05$

(MAD = 1.28). Pearson correlation is $r = 0.89$ ($P < 0.001$), supporting H1.

The linear regression fit is strong:

$$\begin{aligned} S &= 0.94T + 0.82, \\ R^2 &= 0.79, \quad F = 178.6, \quad P < 0.001. \end{aligned} \tag{3}$$

When $T > 8.5$, S rapidly approaches its upper range (above 9.7), suggesting a critical threshold where high-intensity Tang-laugh triggers "sweetness bursts."

The game solution yields a mixed-strategy Nash equilibrium: the defender chooses high defense with probability 62% (low defense 38%); the attacker chooses strong attack with probability 43% (weak attack 57%).

IV. DISCUSSION / DISCUSSION

A. Validity of Robust Statistics

Using MAD and related robust statistics to filter adversarial/outlier ratings reduces the sensitivity of mean-based estimates to extreme values. After filtering, key metrics remain stable within ±0.8, supporting the applicability of robust methods for ACG affect quantification.

B. Robust Regression Addendum: Huber Loss (Optional Enhancement)

To further reduce the influence of extreme clips, we may estimate regression parameters via Huber loss:

$$(\hat{\beta}_0, \hat{\beta}_1) = \arg\min_{\beta_0,\beta_1} \sum_{i=1}^{n} \rho_\delta(S_i - (\beta_0 + \beta_1 T_i)), \tag{4}$$

where

$$\rho_\delta(u) = \begin{cases} \frac{1}{2}u^2, & |u| \le \delta, \\ \delta|u| - \frac{1}{2}\delta^2, & |u| > \delta. \end{cases}$$

C. Cultural Interpretation via Game Theory

The "sweetness–Tang-laugh" attack–defense game reflects affect release and psychological counterbalancing in contemporary online sociality. The combination of "suddenness + sweet facial curvature" efficiently produces sweetness output, aligning with emotional contagion dynamics and explaining the strong meme persistence of Anon' s Tang-laugh in fan communities.

V. CONCLUSION / CONCLUSION

Within a robust-statistics and game-theoretic framework, this study characterizes the linkage between Chihaya Anon' s sweetness and Tang-laugh intensity. Key findings include:

1) Sweetness and Tang-laugh intensity are strongly positively correlated ($r = 0.89$, $P < 0.001$);
2) The defender–attacker game admits a mixed-strategy Nash equilibrium (defender: 62% high defense; attacker: 43% strong attack);
3) When the attack–defense intensity ratio exceeds 1.2, the "break-defense" rate reaches 87.3%, indicating near-total defense failure;
4) Attack–defense characteristics differ across groups, with ages 18–25 forming the core audience of the "sweetness culture."

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